

BAX 423 Homework 2 — Part 1 (Bloom filters)

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StreamEdge Bloom filter setup

Given parameters: $n = 2,000,000$ items, bit array size $m = 19,170,117$, and $k = 6$ independent hash functions. The standard approximate false positive probability when the filter holds n distinct items is

$$p \approx (1 - \exp(-kn/m))^k.$$

(a) False positive probability at capacity

First compute $kn/m = 6 \cdot 2,000,000 / 19,170,117 \approx 0.62587$.

Then $\exp(-kn/m) \approx 0.53474$, so $1 - \exp(-kn/m) \approx 0.46526$, and

$$p \approx (0.46526)^6 \approx 0.010143.$$

So the approximate false positive rate is about **1.01%** (rounded to two decimal places).

(b) Halving p without changing n

To lower the false positive rate you can **increase m** (more bits) and/or **increase k** (more hash functions). Raising k also increases work per lookup, so teams often grow **m** first to hit a target p while keeping k fixed.

Trade-off. Bits are memory. A larger m drives $\exp(-kn/m)$ closer to 1, which shrinks $(1 - \exp(-kn/m))$ and drops p quickly, but RAM at every edge server scales linearly with m .

New m with $k = 6$ fixed. We want $p_{\text{new}} \approx p/2 \approx 0.0050716$. Set $(1 - \exp(-6n/m_{\text{new}}))^6 = p_{\text{new}}$, so $1 - \exp(-6n/m_{\text{new}}) \approx p_{\text{new}}^{1/6} \approx 0.3400$, hence $\exp(-6n/m_{\text{new}}) \approx 0.6600$. Then $6n/m_{\text{new}} \approx -\ln(0.6600) \approx 0.4155$, so $m_{\text{new}} \approx 6n/0.4155 \approx 2.89 \times 10^7$ bits. In other words, about **22.4 million bits** is in the right range to cut the false positive rate roughly in half for this n and k .

(c) Why clearing bits on eviction is wrong

A Bloom filter is a **lossy fingerprint**. Many keys share the same bits. If you flip a bit to 0 because one key was deleted, you may remove evidence for other keys that still map to that bit, which creates **false negatives** (the filter says “definitely not in set” when the item is still cached elsewhere). You cannot safely delete without extra per-cell information.

Structure that supports deletion. A **Counting Bloom filter** stores a small counter per cell instead of a single bit, so inserts increment and deletes decrement (until zero). Alternatives include **cuckoo filters** or other structures designed for approximate membership with safer delete semantics.

(d) Scalable Bloom Filter (SBF)

An SBF is a **sequence of Bloom filters** $B_1, B_2, \dots$. New inserts go into the active filter. When the estimated load would push the false positive rate of the active stage past a budget, you **add a new, larger** Bloom stage rather than rebuilding everything.

Query checks **all** stages (membership is true if any stage returns true). **False positive guarantee:** if each stage i is tuned so its FP probability is at most ϵ_i and the series $\sum \epsilon_i$ is bounded (often geometric), the overall FP probability stays below a chosen cap even as new stages are appended.